Problem Chosen	2027 MCM/ICM Summary Sheet	Team Control Number
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From Model to Evidence: A Reproducible Decay Demonstration

Synthetic Workflow Rehearsal – Demonstration only

Summary

This synthetic example demonstrates how a mathematical assumption becomes a solution, a validation check and a readable figure. We model a remaining amount by exponential decay. Euler and fourth-order Runge–Kutta methods use the same initial state, rate and time grid and are compared with the analytical solution.

At a time step of 0.1, the largest absolute errors over the shared grid are 0.01920100 for Euler and 0.00000033 for Runge–Kutta. These values are generated from the actual demonstration run. We also compare three rate scenarios and propagate 301 specified rate values to a pointwise parameter envelope. The envelope describes assumed parameter variation; it is not a sampling confidence interval.

The example supports an internally checkable numerical comparison and a clear handoff from fixed data to graphics and text. It does not establish the decay equation for any real MCM system, constitute an independent scientific review, or create a human contest approval.

Keywords: exponential decay; numerical validation; sensitivity; parameter uncertainty; reproducible figures.

1 Problem Restatement and Analysis

The demonstration asks three linked questions: how closely two numerical methods reproduce a known solution, how the solution changes with the decay rate, and how an assumed range of rates propagates into an output envelope. Every value is synthetic. Figure 1 connects the model, analytical reference and checks; it is a method diagram rather than evidence about a physical system.

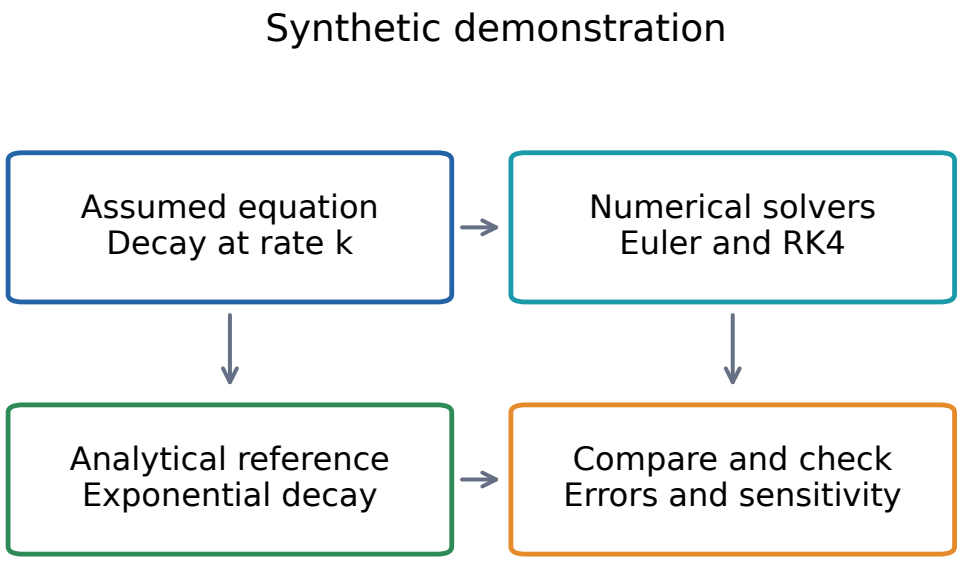


Figure 1: Synthetic method overview: an assumed decay equation, two numerical solvers, an analytical reference and numerical comparison.

2 Assumptions and Symbols

The quantity is dimensionless, its initial value is one, and the rate is constant within each scenario. These are deliberate simplifying assumptions, not measurements. Time is denoted by t, the remaining fraction by $y(t)$, the positive rate by k, and the fixed numerical step by h.

3 Model, Solution and Validation

The shared model and its exact comparison reference are

$$\begin{aligned} \frac{\mathrm{d}y}{\mathrm{d}t} &= -ky, \qquad y(0) = 1, \\ y(t) &= \exp(-kt), \qquad k > 0. \end{aligned} \tag{1}$$

Euler uses $y_{n+1} = y_n - hky_n$; Runge–Kutta combines four intermediate slopes. Both use $h = 0.1$, $k = 1$, and 41 shared time points in $[0, 4]$. We compare both numerical solutions with Equation 1.

3.1 Results and interpretation

The largest absolute grid errors are 0.01920100 for Euler and 0.00000033 for Runge–Kutta. Figure 2 puts both methods on the same comparison scale.

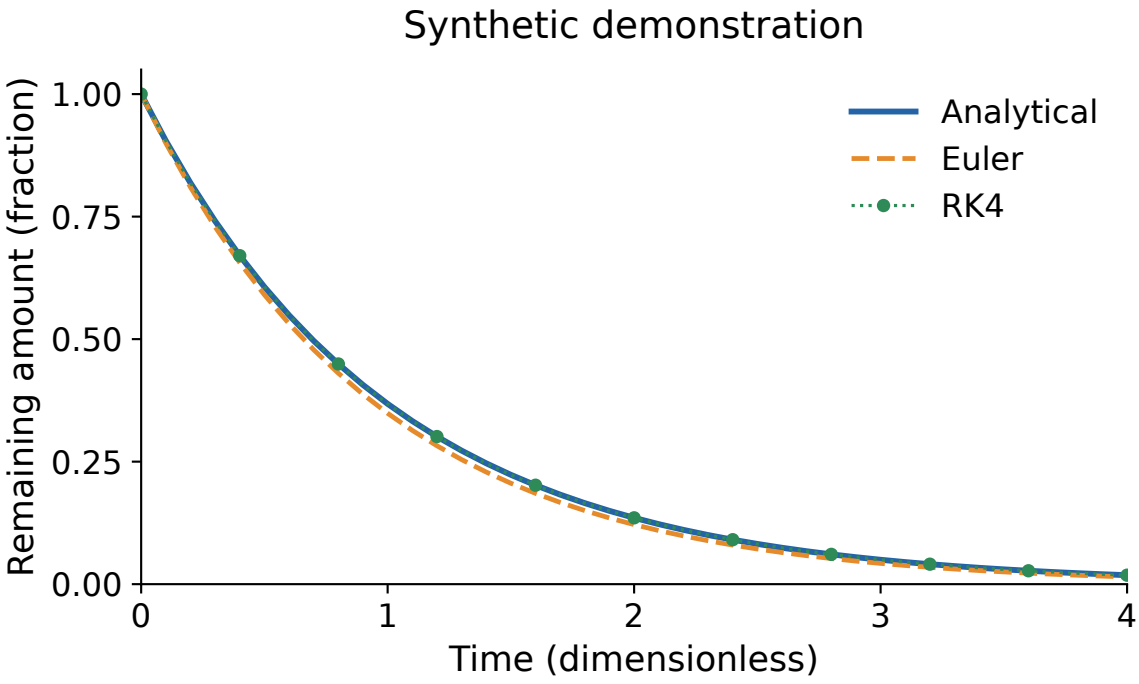


Figure 2: Synthetic numerical comparison. All lines use the same 41 time points; Runge–Kutta markers are shown every fourth point for clarity. Blue: analytical reference; dashed orange: Euler; dotted green: Runge–Kutta. Values and units are dimensionless.

Euler decays too quickly on this grid, while Runge–Kutta follows the exact reference closely. This supports accuracy for this smooth test equation and step; it does not establish accuracy for another model or time scale.

4 Sensitivity and Uncertainty Analysis

4.1 Parameter sensitivity

Figure 3 compares rates of 0.7, 1.0 and 1.3 with identical initial conditions. Increasing the rate lowers the remaining fraction at every positive time. These are scenario responses rather than fits to observations.

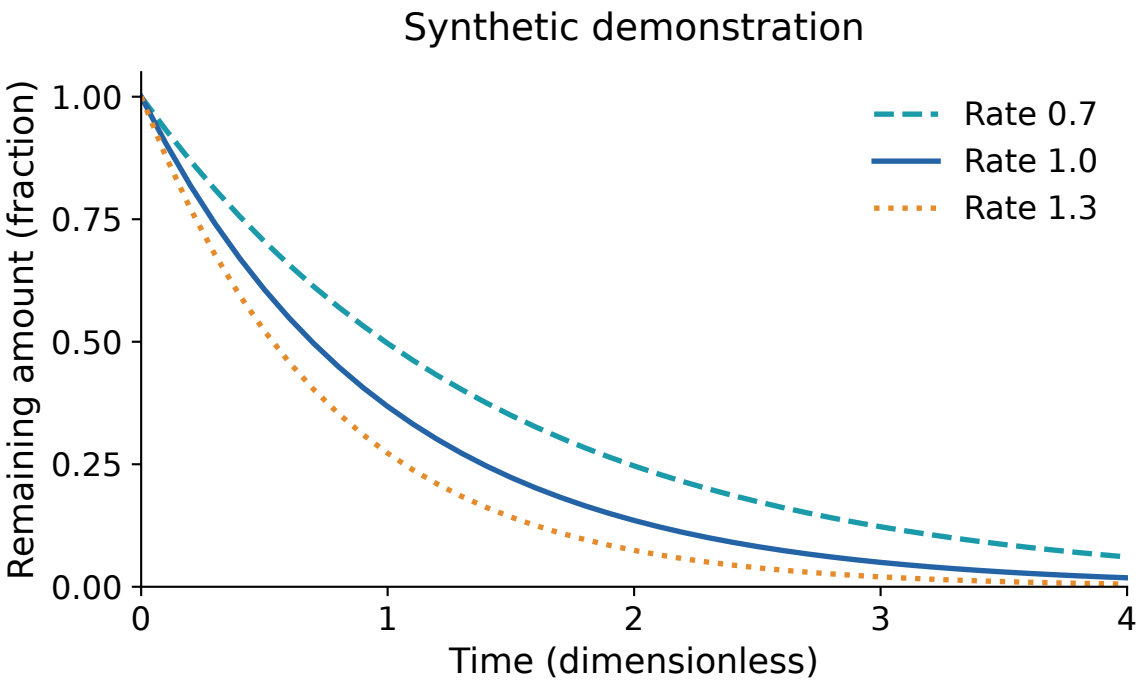


Figure 3: Synthetic sensitivity to the assumed constant decay rate. Line style supplements color so the three scenarios remain distinguishable.

The shared initial value isolates the effect of the rate. The curves separate over time, showing why a model-based decision should report its assumed rate rather than present a single decay curve without qualification.

4.2 A parameter envelope

We evaluate 301 equally spaced rates in [0.7, 1.3] and retain all scenarios. Figure 4 shows the pointwise 10th–90th percentile envelope of those specified rates. It describes this chosen parameter grid, not repeated observations, a fitted posterior or a sampling confidence interval.

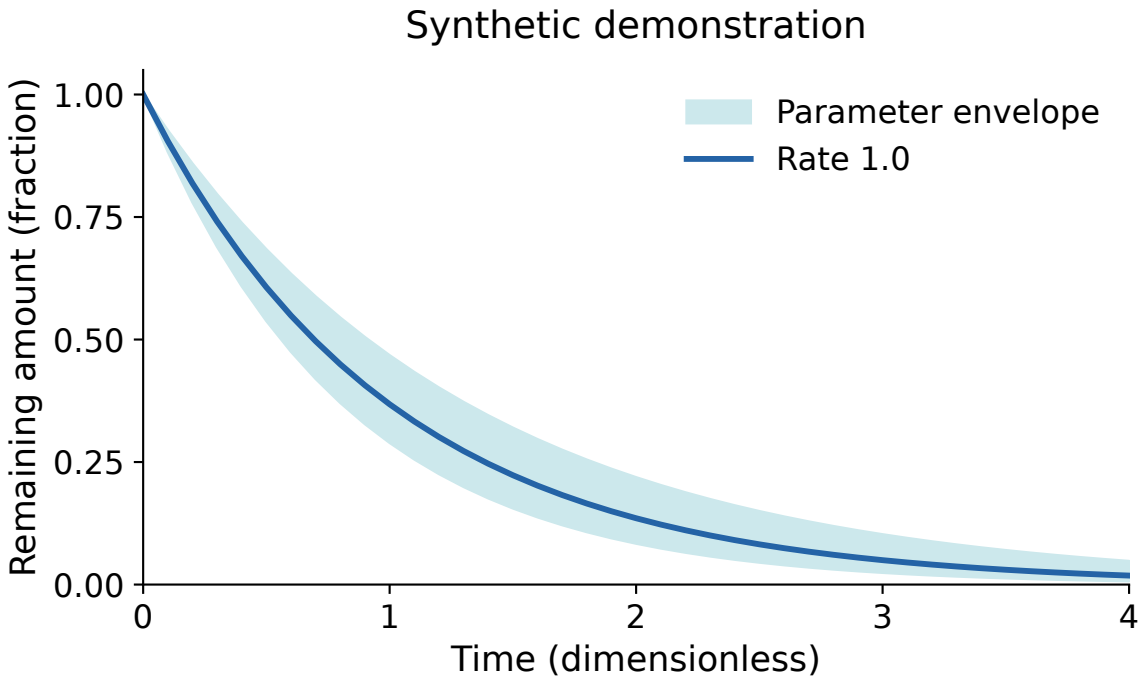


Figure 4: Synthetic propagation of a specified parameter range. Shading is the pointwise 10th–90th percentile across the 301 deterministic rate scenarios; the solid line uses $k = 1$. The band is not a confidence interval.

5 Model Evaluation and Conclusions

The analytical reference, shared grid and exported data make the numerical comparison easy to check. The four graphics have distinct roles: explain the method, compare solvers, assess sensitivity and display uncertainty. The limitations are equally important: there are no observations, no parameter estimation and no independent agent's scientific attestation. Nonconstant rates and noisy measurements would require a new model and appropriate validation. This example demonstrates the reporting pipeline, not the validity of a real-world prediction. Contest presentation and AI disclosure still follow the applicable rules [2, 1].

References

[1] COMAP. Use of AI Tools in COMAP Contests. https://www.contest.comap.org/undergraduate/contests/mcm/flyer/Contest_AI_Policy.pdf, 2025. Version v102025; accessed October 24, 2026.

[2] COMAP. 2027 MCM/ICM Contest Rules, Registration and Instructions. https://www.contest.comap.org/undergraduate/contests/mcm/instructions.php, 2026. Accessed October 24, 2026; verify the applicable rules before competition.

Report on Use of AI Tools

This is a disclosure-structure demonstration, not the AI-use report for an actual competition entry. AI assisted the preparation of the workflow architecture, software, synthetic example and this text.

For a competition project, replace this section using the actual visible conversation records. Identify each tool and model version, describe its purpose, include the required input/output information and document human verification. Unknown model versions must be reported as unknown rather than inferred. The project archive retains separate records for authoring, programming, review and human verification. A successful archive operation does not itself constitute human verification.

The current COMAP policy governs what is disclosed and how it is cited[1]. Missing or partial records must be resolved or clearly reported before the final human release decision.